

2026-06-05 - PWM LED Dimming

Goal

Use Zephyr's PWM API to smoothly dim and brighten an LED using a breathing animation, and understand how to map PWM channels in the Device Tree.

What you will learn

- How to declare a PWM channel in the Device Tree using `pwm-leds`
- How to set duty cycle with `pwm_set_dt`
- How to implement a smooth breathing effect
- How PWM period and pulse width relate to duty cycle and frequency

Overview

PWM (Pulse Width Modulation) rapidly switches a pin on and off. The ratio of on-time (pulse) to total period (period) is the duty cycle. At 1 kHz, the eye and motor cannot follow individual pulses — they perceive an average.

app.overlay

```
/ {
    pwmleds {
        compatible = "pwm-leds";
        pwm_led0: pwm_led_0 {
            pwms = <&pwm0 0 PWM_MSEC(20) PWM_POLARITY_NORMAL>;
            label = "PWM LED";
        };
    };

    aliases {
        pwm-led0 = &pwm_led0;
    };
};

&pwm0 {
    status = "okay";
    pinctrl-0 = <&pwm0_default>;
    pinctrl-names = "default";
};

&pinctrl {
    pwm0_default: pwm0_default {
        group1 {
```

```

        /* LED1 on nRF52840 DK = P0.13 */
        psels = <NRF_PSEL(PWM_OUT0, 0, 13)>;
    };
};
};

```

prj.conf

```

CONFIG_PWM=y
CONFIG_LOG=y

```

Setting duty cycle

```

#include <zephyr/drivers/pwm.h>

static const struct pwm_dt_spec pwm_led =
    PWM_DT_SPEC_GET(DT_ALIAS(pwm_led0));

/* 50% duty cycle */
pwm_set_dt(&pwm_led, pwm_led.period, pwm_led.period / 2);

/* 0% – fully off */
pwm_set_dt(&pwm_led, pwm_led.period, 0);

/* 100% – fully on */
pwm_set_dt(&pwm_led, pwm_led.period, pwm_led.period);

```

Breathing effect

```

/* Ramp pulse from 0 to period in 50 steps, then back down */
uint32_t period = pwm_led.period;
int step = period / 50;

while (1) {
    for (uint32_t pulse = 0; pulse <= period; pulse += step) {
        pwm_set_dt(&pwm_led, period, pulse);
        k_msleep(20);
    }
    for (uint32_t pulse = period; pulse > 0; pulse -= step) {
        pwm_set_dt(&pwm_led, period, pulse);
        k_msleep(20);
    }
}

```

Servo control

PWM is also used for hobby servos (50 Hz, 1–2 ms pulse):

```

/* Neutral position: 1.5 ms pulse at 50 Hz (20 ms period) */
pwm_set_dt(&servo, PWM_MSEC(20), PWM_USEC(1500));

```

```
/* Full left: 1 ms */  
pwm_set_dt(&servo, PWM_MSEC(20), PWM_USEC(1000));  
/* Full right: 2 ms */  
pwm_set_dt(&servo, PWM_MSEC(20), PWM_USEC(2000));
```

Why this matters

PWM is used everywhere: LED brightness control, motor speed, servo position, buzzer tones, and switching power supplies. Zephyr's Device Tree mapping means you change a pin number in the overlay, not in the application code.

Practice tasks

1. Flash and observe the breathing LED on LED1 (P0.13).
2. Change the step size and sleep delay to speed up or slow down the effect.
3. Wire a buzzer to a PWM pin and play a 1 kHz tone.

Example folder

See `src/2026-06-05_PWM_LED_Dimming` for the day's code.

Next topic

Tomorrow we store configuration data in internal flash using Zephyr's NVS.